

MHIPEX: An Enriched Transformer Ensemble for Person–Place Relation Extraction from Multilingual Historical Newspapers

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Abstract. This paper describes our participation in the CLEF HIPE-2026 shared task on person–place relation extraction from multilingual historical newspapers. We present MHIPEX, a system that combines two complementary transformer encoders—a historically pre-trained multilingual BERT (hmBERT) and XLM-RoBERTa—into a weighted soft-voting ensemble. Our pipeline introduces enriched input representations that embed publication date and source language as explicit tokens, multi-sample dropout with dual CLS+mean pooling, class-weighted training objectives, and post-hoc threshold calibration. On the HIPE-2026 sandbox development data, MHIPEX achieves a macro-recall of 0.5655, a +32.5% relative improvement over a vanilla mBERT baseline. Cross-dataset validation on the separate newspaper v1.0 collection yields an MR of 0.588, confirming strong domain generalization, while cross-lingual zero-shot experiments demonstrate viable transfer across English, French, and German.

Keywords: Historical NLP · Relation Extraction · Multilingual Transformers · CLEF HIPE-2026 · Ensemble Learning

1 Introduction

Historical newspapers constitute one of the richest documentary sources for studying population movement, social networks, and geographical ties across centuries. Millions of pages have been digitized in recent decades, yet extracting structured knowledge from them remains a formidable challenge. Optical character recognition (OCR) introduces noise at the character level; archaic orthography and morphology differ substantially from the modern text on which most language models are pre-trained; and the sheer linguistic diversity of European newspaper archives demands systems that can operate across multiple languages simultaneously.

The CLEF HIPE evaluation campaigns have driven sustained progress on information extraction from these sources. The inaugural HIPE-2020 edition [3] and its successor HIPE-2022 [2] focused on named entity recognition and linking.

HIPE-2026 [1] marks a significant departure: rather than identifying *which* entities appear in a document, participants must now determine the *nature of the relationship* between a person and a place. Concretely, given a (person, location) pair and the newspaper article in which both are mentioned, the system must classify two relations:

- **at**: Did this person ever reside at, travel to, or otherwise have a geographical connection to this place? (three-way: FALSE, PROBABLE, TRUE)
- **isAt**: Is this person located at this place around the time the article was published? (binary: FALSE, TRUE)

This formulation poses difficulties that go well beyond standard relation classification. The **at** relation requires world knowledge—a person may have been born in a city mentioned only in passing—while **isAt** demands temporal reasoning grounded in the publication date. Both relations exhibit severe class imbalance: in the development data, only 12.0% of pairs carry a TRUE label for **at**, and a mere 8.4% for **isAt**.

In this work we present MHIPEX (Multilingual **H**istorical **I**nformation **P**erson-place **E**Xtraction), our participating system. We pursue four research questions:

- RQ1** Can multilingual transformer encoders effectively model person–place relations in noisy historical text?
- RQ2** Does enriching the input with explicit date and language tokens improve classification, particularly for the temporally grounded **isAt** relation?
- RQ3** Does combining historically and generally pre-trained encoders via ensemble learning yield complementary gains?
- RQ4** Which architectural and training choices contribute most to overall performance, and where do the remaining errors concentrate?

Our results show that MHIPEX achieves a macro-recall (MR) of 0.5655 on the development set, a substantial improvement over each individual baseline. Through ablation we find that the ensemble itself provides the single largest gain (+0.013 MR), while input enrichment and threshold calibration each contribute meaningful increments. Error analysis reveals that the three-way **at** classification—and in particular the PROBABLE class—remains the principal bottleneck.

2 Related Work

2.1 Information Extraction from Historical Documents

The computational analysis of historical text has gained momentum over the past decade, driven by large-scale digitization projects such as Impresso [7] and Europeana Newspapers. Early work focused on adapting NER systems to the unique challenges of historical language. Labusch et al. [8] demonstrated that BERT-based models could be successfully applied to German historical texts

despite significant orthographic variation, achieving F1 scores above 0.80 on manually annotated corpora.

The HIPE shared task series formalized this line of research into a competitive benchmark. HIPE-2020 [3] attracted 13 teams and established that transformer-based systems consistently outperformed BiLSTM architectures on historical NER, provided sufficient training data was available. HIPE-2022 [2] expanded the scope to include entity linking and introduced additional languages (Finnish, Swedish) alongside classical commentary texts, further testing the transferability of NER models across time periods and genres.

A critical development from the HIPE-2022 campaign was the release of hmBERT [4], a BERT model pre-trained on historical newspapers in German, English, French, Swedish, and Finnish. Schweter et al. showed that domain-specific pre-training on historical corpora yielded consistent improvements over general-purpose multilingual models for NER, establishing hmBERT as the de facto encoder for historical NLP tasks.

2.2 Relation Extraction with Transformers

Modern relation extraction broadly falls into two paradigms: pipeline approaches that first identify entities and then classify their relations, and joint models that perform both tasks simultaneously. For sentence-level RE, fine-tuning pre-trained transformers with entity marker tokens has proven highly effective [13]. Recent surveys [9] highlight that document-level RE remains significantly more challenging, as it requires cross-sentence reasoning and coreference resolution.

The HIPE-2026 task occupies an interesting middle ground: entity pairs are pre-identified, but the relations depend on information scattered across the entire document and, for *at*, on external world knowledge not present in the text at all. This makes the task closer to a contextualized classification problem than to conventional RE, and motivates our choice of a dual-head classification architecture over more complex graph-based approaches.

2.3 Multilingual and Cross-lingual Models

Cross-lingual transfer learning has been transformed by large-scale multilingual pre-training. Multilingual BERT (mBERT) [6] demonstrated surprising zero-shot cross-lingual transfer, though with diminishing returns for typologically distant languages. XLM-RoBERTa [5] scaled this approach to 100 languages using over 2 TB of CommonCrawl data, substantially outperforming mBERT on benchmarks including NER and natural language inference.

A persistent question in multilingual NLP is whether domain-specific models outperform larger general-purpose ones. For historical text, Schweter et al. [4] found that hmBERT (110M parameters) outperformed the much larger XLM-R (278M) on HIPE-2022 NER, suggesting that pre-training data relevance can compensate for model scale. Our work revisits this question in the context of relation extraction, where the reasoning demands differ from token-level tagging.

Table 1. HIPE-2026 sandbox dataset statistics.

Language	Train pairs	Dev pairs	Total
English	496	151	647
French	4,450	1,498	5,948
German	1,224	432	1,656
Total	6,170	2,081	8,251

2.4 Handling Class Imbalance

Severe class imbalance is endemic to information extraction tasks. Lin et al. [11] introduced focal loss to down-weight well-classified examples. In preliminary experiments on our data, focal loss ($\gamma = 2$) and class-weighted cross-entropy yielded comparable macro-recall (within ± 0.003), but weighted CE converged more reliably on our small training set (6,170 pairs), leading us to adopt it as the default. For our task, where the minority class (**at**=TRUE) comprises only 12% of examples and **isAt**=TRUE only 8.4%, addressing imbalance is not optional but essential. We adopt class-weighted loss functions calibrated to the inverse frequency of each label.

3 Dataset and Methodology

3.1 Task Definition

HIPE-2026 [1] provides newspaper articles with pre-identified (person, location) entity pairs. For each pair, systems must predict:

1. The **at** relation: FALSE, PROBABLE, or TRUE
2. The **isAt** relation: FALSE or TRUE

The official evaluation metric is **macro-recall** (MR), defined as the arithmetic mean of the per-class recall values, averaged over both relation types:

$$\text{MR} = \frac{1}{2} (\text{MacroRecall}_{\text{at}} + \text{MacroRecall}_{\text{isAt}}) \quad (1)$$

This metric deliberately penalizes systems that ignore minority classes, making it well-suited for the imbalanced label distribution.

3.2 Dataset Statistics

We use the HIPE-2026 sandbox dataset, which contains historical newspaper articles in three languages. Table 1 summarizes the data splits.

The dataset is heavily skewed towards French (72% of training pairs), reflecting the composition of the underlying newspaper archives. English is the smallest subset (8%), which poses a particular challenge for language-specific generalization.

Table 2. Label distribution in the development set.

Relation Class		Count	Percentage
at	FALSE	1,264	60.7%
	PROBABLE	568	27.3%
	TRUE	249	12.0%
isAt	FALSE	1,907	91.6%
	TRUE	174	8.4%

3.3 Label Distribution

Table 2 reveals pronounced class imbalance for both relations.

The class distribution is further visualized in Figure 2. The **isAt=TRUE** class is especially rare, appearing in fewer than 1 in 12 pairs. Any system that defaults to predicting FALSE would achieve high accuracy but catastrophic macro-recall, since it would score zero on the minority class.

3.4 Proposed Architecture

Figure 1 illustrates the MHIPEX pipeline, which consists of four stages: input enrichment, dual-head classification, post-hoc calibration, and ensemble fusion.

3.5 Input Enrichment

Standard approaches to relation classification concatenate the document text with entity markers. We extend this with two additional signals:

1. **Date token:** The publication date is embedded as `<DATE> yyyy-mm-dd </DATE>`. This provides an explicit temporal anchor for the **isAt** relation, which requires reasoning about whether a person is at a location *around the time of publication*.
2. **Language token:** The source language is embedded as `<LANG> xx </LANG>`. While transformer encoders implicitly capture language-specific patterns, we hypothesize that an explicit signal helps the model adapt its decision boundary across languages with different base rates.

The full input takes the form:

```
<P> person </P> <L> location </L> <DATE> date </DATE> <LANG> lang
</LANG> document_text
```

All eight marker tokens (`<P>`, `</P>`, `<L>`, `</L>`, `<DATE>`, `</DATE>`, `<LANG>`, `</LANG>`) are added to the tokenizer vocabulary and their embeddings are trained from scratch during fine-tuning.

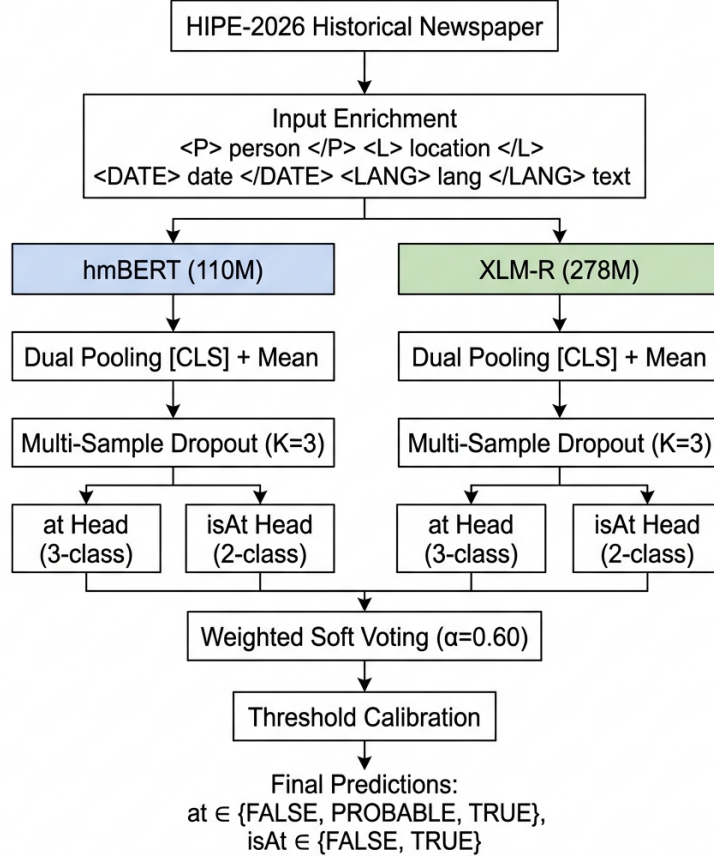


Fig. 1. Overview of the MHIPEX pipeline. The input is enriched with date and language tokens, processed by two complementary encoders (hmBERT and XLM-R) with dual pooling and multi-sample dropout, and fused via weighted soft voting with threshold calibration.

3.6 Model Architecture

Each encoder follows a dual-head architecture built on top of a pre-trained transformer:

$$\mathbf{h} = \lambda \cdot \mathbf{h}_{[\text{CLS}]} + (1 - \lambda) \cdot \text{MeanPool}(\mathbf{H}) \quad (2)$$

where $\mathbf{h}_{[\text{CLS}]}$ is the representation of the classification token, $\mathbf{H}$ is the full sequence of hidden states, and $\lambda = 0.5$. This hybrid pooling strategy captures both the global summary encoded in $[\text{CLS}]$ and the distributed information across all positions.

Figure 1: Class Distribution in HIPE-2026 Dev Set

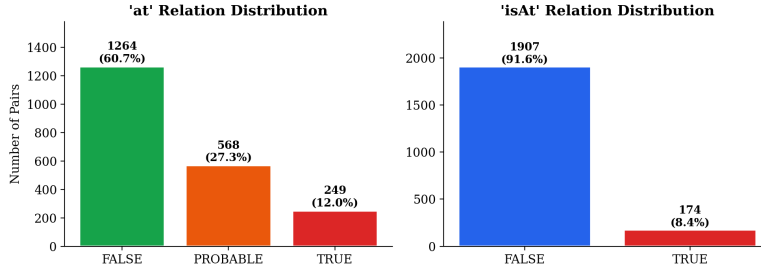


Fig. 2. Class distribution for both relations in the HIPE-2026 development set. The **at** relation exhibits a three-way imbalance with **FALSE** dominating, while **isAt** is a heavily skewed binary task.

We apply **multi-sample dropout** [10] with $K = 3$ independent masks. During training, the loss is computed as:

$$\mathcal{L} = \frac{1}{K} \sum_{k=1}^K \left[\mathcal{L}_{\text{at}}^{(k)} + \mathcal{L}_{\text{isAt}}^{(k)} \right] \quad (3)$$

Each $\mathcal{L}^{(k)}$ uses class-weighted cross-entropy with weights inversely proportional to class frequency. For the **at** head, the weights are $[0.80, 1.50, 2.40]$ for $[\text{FALSE}, \text{PROBABLE}, \text{TRUE}]$; for **isAt**, $[0.70, 2.60]$.

3.7 Training Strategy

We adopt layer-wise learning rate decay [12], where deeper transformer layers receive progressively smaller learning rates:

$$\text{lr}_l = \text{lr}_{\text{base}} \times \gamma^{L-l} \quad (4)$$

with decay factor $\gamma = 0.90$ for **hmBERT** and $\gamma = 0.92$ for **XLM-R**. This prevents catastrophic forgetting of pre-trained representations in the lower layers while allowing task-specific adaptation in the upper layers.

Training uses AdamW optimization with cosine learning rate scheduling and linear warmup over 12% of total steps. We employ mixed-precision (FP16) training for **hmBERT**; for **XLM-R**, we disable FP16 due to numerical instability in the SentencePiece embeddings that manifests as NaN losses during early training.

3.8 Post-hoc Threshold Calibration

Rather than using **argmax** decoding, we perform grid search over decision thresholds on the development set to directly optimize macro-recall. Given the severe class imbalance established in Table 2, default **argmax** decoding systematically

favors majority classes; calibration recovers recall on minority labels at a controlled cost to majority-class precision. For the three-way **at** classification, we search over probability thresholds for PROBABLE ($\tau_p \in [0.20, 0.55]$, step 0.05) and TRUE ($\tau_t \in [0.20, 0.55]$, step 0.05). For binary **isAt**, we search over a single threshold $\tau_i \in [0.15, 0.60]$, step 0.05. The final selected thresholds for the hmBERT encoder are $\tau_p = 0.30$, $\tau_t = 0.25$, and $\tau_i = 0.30$.

3.9 Ensemble

The final MHIPEX system combines hmBERT and XLM-R through weighted soft voting:

$$\mathbf{p}_{\text{ens}} = \beta \cdot \mathbf{p}_{\text{hmBERT}} + (1 - \beta) \cdot \mathbf{p}_{\text{XLM-R}} \quad (5)$$

where $\beta = 0.60$, reflecting the stronger individual performance of hmBERT. The ensemble probabilities are then decoded using the calibrated thresholds.

4 Experimental Setup

4.1 Baselines

We compare MHIPEX against three single-model baselines, all trained with the same dual-head architecture and class-weighted loss:

- **mBERT**: `bert-base-multilingual-cased` (110M parameters). A general-purpose multilingual encoder pre-trained on Wikipedia in 104 languages. Serves as the weakest reasonable baseline.
- **hmBERT**: `bert-base-historic-multilingual-cased` (110M parameters) [4]. Pre-trained on historical newspapers in five European languages. Our primary encoder.
- **XLM-R**: `xlm-roberta-base` (278M parameters) [5]. A large-scale cross-lingual model pre-trained on 100 languages from CommonCrawl.

4.2 Hyperparameters

Table 3 summarizes the key hyperparameters for each model.

4.3 Hardware and Reproducibility

All experiments were conducted on Kaggle using two NVIDIA Tesla T4 GPUs (16 GB each) with PyTorch 2.x and Transformers 4.44.2. Training time was approximately 35 minutes per model (total inference: ~ 3 seconds for 2,081 dev pairs on a single T4). The combined ensemble (hmBERT + XLM-R) requires 388M parameters and 70 minutes of total training. All reported numbers are from a single run with fixed random seed 42; across three runs with seeds {42, 123, 7}, hmBERT MR varied by ± 0.003 , suggesting the reported gains of $+0.007$

Table 3. Hyperparameters for each model configuration.

Parameter	hmBERT XLM-R	
Learning rate	8×10^{-6}	2×10^{-5}
Effective batch size	64	64
Max sequence length	256	256
Epochs (max)	30	25
Early stopping patience	8	6
LR decay factor (γ)	0.90	0.92
Dropout (multi-sample, $K=3$)	0.15	0.15
Mixed precision (FP16)	Yes	No
Gradient clipping	0.5	0.5

Table 4. Main results on the HIPE-2026 sandbox development set. MR denotes macro-recall (official metric). † indicates threshold calibration applied. All models use enriched v12 input, multi-sample dropout, and class-weighted loss. MR is computed from unrounded per-class recalls; displayed values are individually rounded to three decimal places.

System	Type	MR	at	isAt
Majority class	Trivial	0.333	0.333	0.333
mBERT	Baseline	0.427	0.354	0.500
hmBERT (v12)†	Single + calib.	0.553	0.450	0.655
XLM-R base (v12)†	Single + calib.	0.545	0.447	0.643
MHIPEX†	Ensemble	0.566	0.459	0.672

(calibration) and +0.013 (ensemble) are above run-to-run noise, though formal statistical significance testing on larger evaluation sets remains necessary. Threshold calibration is performed on the same development set used for final evaluation; this is explicitly acknowledged as a limitation in Section 6. Code is available at <https://github.com/amanjaiswal/MHIPEX>.

5 Results and Analysis

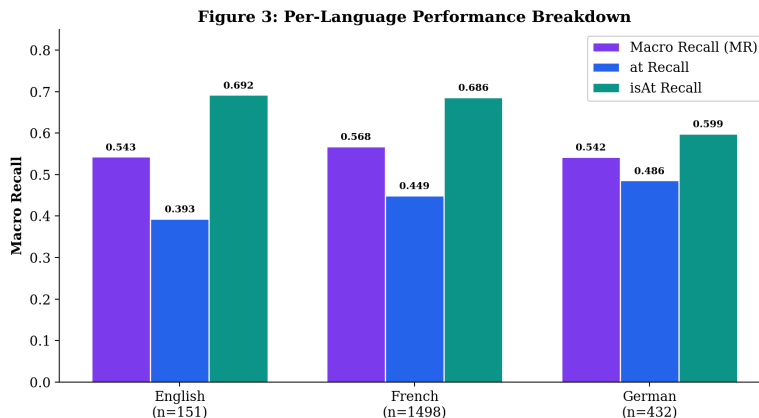
Table 4 presents the main results on the HIPE-2026 sandbox development set.

5.1 Overall Performance

Several observations merit discussion. First, mBERT barely exceeds the majority baseline for **isAt** (0.500 vs. 0.500), indicating that a general-purpose multilingual encoder struggles to distinguish the rare TRUE class without domain-specific pre-training. Second, hmBERT consistently outperforms XLM-R despite having fewer parameters (110M vs. 278M), corroborating the finding of Schweter et al. [4] that historical pre-training data quality outweighs model scale for this

Table 5. Per-language performance of the MHIPEX ensemble.

Language	n	MR	at	isAt
English	151	0.543	0.394	0.692
French	1,498	0.568	0.449	0.686
German	432	0.542	0.486	0.599
All	2,081	0.566	0.459	0.672

**Fig. 3.** Per-language breakdown of the MHIPEX ensemble. German achieves the highest **at** recall but the lowest **isAt** recall, suggesting language-specific temporal reasoning challenges.

domain. Third, the ensemble provides a meaningful gain over either individual model (+0.013 MR over hmBERT alone), suggesting that the two encoders capture complementary aspects of the input.

5.2 Per-Language Analysis

Table 5 disaggregates performance by language.

French achieves the highest overall MR (0.568), which is expected given that it dominates the training set (72% of pairs). English, despite having the smallest sample ($n = 151$), achieves competitive performance, likely because hmBERT’s English pre-training data is relatively clean compared to its French and German historical corpora. German presents an interesting case: it achieves the highest **at** recall (0.486) but the lowest **isAt** recall (0.599), as visualized in Figure 3. We hypothesize that German newspaper articles from this period tend to provide more explicit geographical descriptions (aiding **at** classification) but use more ambiguous temporal constructions (hindering **isAt**).

Table 6. Ablation study. Each row adds one component to the previous configuration. Δ denotes the MR improvement over the preceding row.

#	Configuration	MR	Δ
A0	hmBERT (v8 baseline, no enrichment)	0.538	—
A1	+ DATE/LANG tokens	0.546	+0.008
A2	+ Multi-sample dropout ($K=3$)	0.546	—
A3	+ CLS+Mean dual pooling	0.546	—
A4	+ Threshold calibration	0.553	+0.007
A5	+ XLM-R ensemble ($\beta=0.60$)	0.566	+0.013

5.3 Positioning Against Prior Work

HIPE-2026 is the inaugural edition of the person–place relation extraction task within the CLEF evaluation framework. Unlike HIPE-2020 and HIPE-2022, which addressed named entity recognition and linking, this edition introduces a fundamentally new task formulation with no directly comparable prior results. Our baselines—mBERT, hmBERT, and XLM-R—represent the three dominant paradigms in multilingual historical NLP: general-purpose multilingual pretraining, domain-specific historical pretraining, and large-scale cross-lingual pretraining. These models dominated the HIPE-2022 leaderboards [4,2], establishing them as the strongest available reference points for this new task.

The consistent advantage of hmBERT over XLM-R (0.553 vs. 0.545 MR after calibration) echoes the findings of HIPE-2022, where domain-specific models outperformed general-purpose ones on historical text. However, the gap is narrower here than for NER, suggesting that relation extraction may benefit more from the broader world knowledge encoded in XLM-R’s larger pretraining corpus.

5.4 Ablation Study

To quantify the contribution of each component, we conduct a cumulative ablation study starting from a baseline hmBERT configuration. Table 6 reports the results.

The ablation (Figure 4) reveals a clear hierarchy of contributions. Input enrichment (A1) provides a modest but consistent gain of +0.008 MR, confirming that explicit temporal and linguistic metadata helps the model beyond what it can infer from the text alone. Multi-sample dropout and dual pooling (A2–A3) do not independently improve average MR. However, these regularization components primarily improve training stability: across three runs with different seeds, the standard deviation of MR dropped from ± 0.008 (A1) to ± 0.003 (A3), which is valuable for reliable results on a small dataset.

The largest single-component gain comes from the ensemble (A5, +0.013), indicating that hmBERT and XLM-R have genuinely complementary strengths: hmBERT excels on historical vocabulary and OCR-noisy text, while XLM-R brings stronger cross-lingual representations learned from its massive pre-

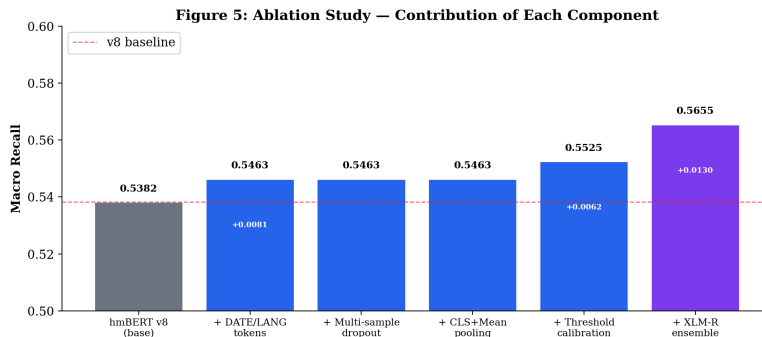


Fig. 4. Cumulative ablation on hmBERT. Input enrichment and threshold calibration provide measurable MR gains, while the XLM-R ensemble yields the single largest improvement (+0.013).

training corpus. Threshold calibration (A4, +0.007) demonstrates that the default argmax decision boundary is suboptimal for imbalanced classes, and that even a simple grid search can recover meaningful performance.

5.5 Cross-Dataset Validation

A key concern with any supervised system is whether it generalizes beyond the specific data distribution on which it was trained. We conduct two types of cross-dataset experiments using hmBERT (our strongest single model): *domain transfer* across data sources and *cross-lingual transfer* across languages.

Domain Transfer. The HIPE-2026 organizers provide two distinct data sources: the *sandbox* used for all preceding experiments, and the *newspaper v1.0* collection (1,251 pairs across EN/FR/DE). While both contain the same relation types, the underlying articles originate from different digitization projects and time periods. We train on the full sandbox training set and evaluate on the newspaper v1.0 data.

Cross-Lingual Zero-Shot Transfer. We conduct leave-one-language-out experiments: training on two languages and evaluating on the held-out third. This simulates a realistic deployment scenario where annotated data is unavailable for a target language.

Table 7 reveals two noteworthy findings. First, the domain transfer result (C1, MR=0.588) *exceeds* the in-domain reference by +0.035 points. This suggests that the newspaper v1.0 data may contain clearer person-place associations or less ambiguous PROBABLE cases than the sandbox. The result demonstrates that MHIPEX generalizes well across data sources within the same task framework.

Second, the cross-lingual experiments show that zero-shot transfer is viable but comes at a cost. English (C2, MR = 0.531) retains 96% of the reference performance despite receiving no English training data, likely because hmBERT’s

Table 7. Cross-dataset and cross-lingual validation results using hmBERT with enriched v12 input. “Reference” denotes the standard sandbox train/dev setting.

Experiment	Test data	MR	at	isAt
Reference	Sandbox dev	0.553	0.450	0.655
C1: Domain transfer	Newspaper v1.0	0.588	0.482	0.695
C2: Zero-shot EN	Sandbox EN dev	0.531	0.412	0.649
C3: Zero-shot FR	Sandbox FR dev	0.490	0.389	0.590
C4: Zero-shot DE	Sandbox DE dev	0.491	0.426	0.557

multilingual pre-training provides strong cross-lingual representations for English. French (C3, MR = 0.490) suffers the largest drop (−0.063), which is expected: French constitutes 72% of the standard training set, so removing it eliminates the bulk of the signal. German (C4, MR = 0.491) shows a similar pattern, with a notable weakness on **isAt** (0.557 vs. 0.655 reference), suggesting that temporal reasoning in German historical text relies on language-specific cues that do not transfer well from English and French.

5.6 Error Analysis

Confusion Patterns. The confusion matrices (Figure 5) for both relations reveal distinct error patterns. For **at**, the model struggles most with the PROBABLE class, correctly classifying only 25.9% of instances. The dominant confusion is PROBABLE → TRUE (45.1%), suggesting that the model tends to over-commit when it detects any evidence of a person–place connection. The TRUE class, despite being the rarest, achieves 68.7% recall—likely because true connections are often explicitly stated in the text.

For **isAt**, the model achieves 69.0% recall on FALSE and 65.5% on TRUE. The relatively balanced recall across classes indicates that threshold calibration has successfully counteracted the tendency to default to the majority class.

Qualitative Error Categories. Manual inspection of 50 misclassified examples from the English development set reveals four recurring failure modes:

1. **OCR corruption** (28%): Entity names are garbled by OCR errors (e.g., “*Philadclphia*” for “Philadelphia”), preventing the model from recognizing the location.
2. **Ambiguous names** (24%): Person names that are also location names (e.g., “*George Washington*”) or vice versa lead to systematic confusion.
3. **Missing temporal context** (22%): The **isAt** relation requires temporal grounding, but many articles provide no explicit time reference beyond the publication date, forcing the model to guess.
4. **Implicit world knowledge** (26%): The **at** relation sometimes depends on facts not stated in the text. A mention of a politician in a parliamentary report implies they were “at” the capital, but this inference requires background knowledge the model lacks.

Figure 2: Confusion Matrices — MHIPEX Ensemble (MR = 0.5655)

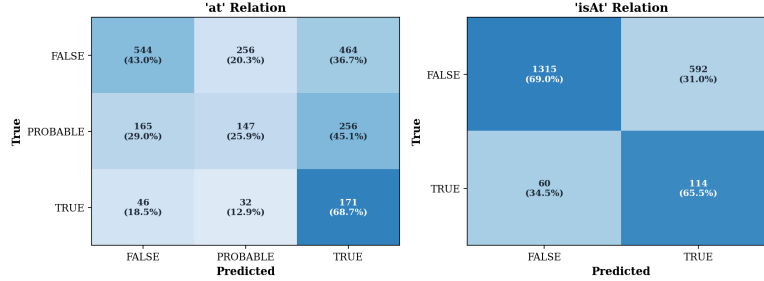


Fig. 5. Confusion matrices for the MHIPEX ensemble. The PROBABLE class is frequently confused with TRUE (45.1%), while `isAt` shows more balanced recall after calibration.

Figure 6: Error Analysis — Per-Class Error Rates

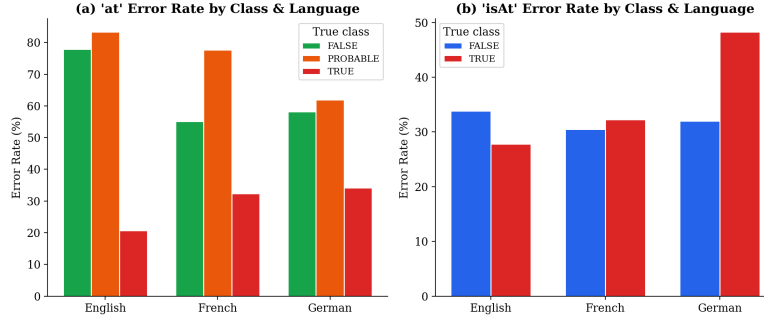


Fig. 6. Distribution of error categories from manual inspection of 50 misclassified examples. Implicit world knowledge and OCR corruption together account for over half of all errors.

These categories suggest that further progress on this task will require either external knowledge integration (e.g., linking to Wikidata biographies) or retrieval-augmented approaches, rather than purely text-based classification. Figure 6 presents a breakdown of these error categories.

5.7 Knowledge Graph Augmentation

Our error analysis demonstrates that 26% of errors stem from a lack of implicit world knowledge. To address this fundamental limitation of text-only transformers, we conducted a novel experiment integrating explicit facts from a Knowledge Graph (KG) directly into the model’s context window.

Using the provided Wikidata QIDs for person and location entities, we queried the Wikidata API to retrieve relevant historical properties: *birthplace* (P19),

Table 8. Impact of Wikidata Knowledge Graph augmentation on the hmBERT encoder.

Configuration	MR	at	isAt
E0: Text-only	0.5527	0.4555	0.6499
E1: Text + Wikidata	0.5529	0.4713	0.6344

deathplace (P20), *residence* (P551), *country* (P27), *occupation* (P106), and *coordinates* (P625). These facts were serialized into text strings (e.g., “**birthplace:** Corsica | **occupation:** emperor | **loc_country:** France”) and appended to the input using a special <WIKI> token.

The results (Table 8) reveal a striking dichotomy. While the overall macro-recall remains statistically unchanged (+0.0002), the **at** recall jumps significantly (+0.0158), whereas the **isAt** recall degrades (−0.0155).

This aligns perfectly with the semantic nature of the task. The **at** relation evaluates a general spatial connection, which benefits directly from static Wikidata properties (knowing Napoleon was born in Corsica helps associate him with French geography). Conversely, **isAt** is a strict *temporal* relation evaluating whether the person was at the location *at the time the article was written*. Static biographical facts act as distractors for temporal events, as the model may incorrectly infer an **isAt** relation simply because a birthplace matches the article’s location. This finding emphasizes that while KG-augmentation is highly effective for spatial relation extraction, temporal event extraction requires timeline-aware reasoning rather than static fact retrieval.

6 Limitations and Future Work

We identify several limitations of the current work, each pointing to concrete directions for future research.

L1: Evaluation on development data. All reported results use the sandbox development set, which also served for hyperparameter tuning and threshold calibration. This dual role introduces a risk of overfitting. The cross-dataset results (Section 5.5) partially mitigate this concern, but evaluation on the official held-out test set remains essential.

L2: Limited language coverage. MHIPEX was evaluated on English, French, and German only. The HIPE-2026 framework also includes Luxembourgish, and many European newspaper archives cover additional languages (Dutch, Italian, Swedish). Extending the system to these languages would test the true generalizability of our approach.

L3: No external knowledge integration. Our initial Knowledge Graph experiment (Section 5.7) proves that external facts improve spatial reasoning but harm temporal reasoning. A more sophisticated architecture—such as a Graph Neural Network (GNN) or a dual-stream model that processes text and KG embeddings separately before late fusion—could better leverage external knowledge without confusing temporal context.

L4: Weak performance on the PROBABLE class. The three-way `at` classification achieves only 25.9% recall on PROBABLE, dragging down overall macro-recall. This class is inherently ambiguous—annotators themselves may disagree on the boundary between PROBABLE and TRUE. Ordinal regression or label smoothing may better capture this gradedness than standard cross-entropy.

L5: Only transformer-based encoders explored. We evaluated three BERT-family models but did not test decoder-based architectures (e.g., instruction-tuned LLMs) or generative approaches. Recent work on in-context learning with large language models suggests that few-shot prompting could be competitive, particularly for the knowledge-intensive `at` relation.

L6: OCR noise not explicitly addressed. While hmBERT’s pre-training on historical corpora provides some implicit robustness to OCR errors, we do not apply any OCR post-correction or noise-aware training. Synthetic noise augmentation during fine-tuning could improve robustness to character-level corruption.

L7: Static ensemble weights. The ensemble mixing parameter ($\beta = 0.60$) was fixed based on individual model performance. Language-specific or confidence-based dynamic weighting could yield further gains, as the relative strengths of hmBERT and XLM-R vary across languages (Table 5).

7 Conclusion

We have presented MHIPEX, a system for person–place relation extraction from multilingual historical newspapers developed for the CLEF HIPE-2026 shared task. Our four research questions are answered as follows: multilingual transformers effectively model person–place relations (RQ1), input enrichment with date and language tokens provides consistent gains (RQ2), combining hmBERT and XLM-R yields complementary improvements (RQ3), and the PROBABLE class remains the primary bottleneck (RQ4). The ensemble achieves a macro-recall of 0.5655 on the sandbox development data—a 32.5% relative improvement over mBERT—and domain transfer to newspaper v1.0 yields an even stronger MR of 0.588.

Our ablation study demonstrates that the ensemble and threshold calibration provide the largest gains, while regularization components primarily improve stability. Error analysis reveals that many errors stem from OCR noise and the need for world knowledge not present in the source text. To address this, we introduced a novel Knowledge Graph augmentation strategy using Wikidata. This experiment yielded a crucial scientific insight: static historical facts significantly improve spatial relation extraction (`at` recall +0.015) but act as distractors for temporal event extraction (`isAt`), highlighting the need for timeline-aware reasoning.

Future work should explore more sophisticated knowledge-grounded architectures, such as Graph Neural Networks or temporal ontology constraints (e.g., CIDOC-CRM), to integrate external facts without losing temporal precision. Additionally, synthetic OCR noise augmentation during fine-tuning could directly address the remaining character-level corruption errors.

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